EIE1005 Fundamental AI and Data Analytics

AI and Data Analytics in the Virtual World



Richard Pang

Tel: 27664745
Email: richard.pang@polyu.edu.hk

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References

- G.N. Yannakakis and J. Togelius, *Artificial intelligence and games*, Springer, 2018
- T. Simonini and O. Saneviero, *An Introduction to Deep Reinforcement Learning Class,* https://github.com/huggingface/deep-rl-class
- A. Majumder, *Deep Reinforcement Learning in Unity: With Unity ML Toolkit*. Berkeley, CA: Apress, 2020.
- D. Buckley, “Unity ML-Agents Tutorials – Complete Guide,” *GameDev Academy*, 2022 (online resource).
- T. Simonini, “An Introduction to Unity ML-Agents,” Towards Data Science, 2020 (online resource)

Part I

Development of game AI

What is Game AI?

- Imitate intelligence in the actions of non-player characters (NPCs) and procedural content generation (PCG)
- Make the game “feel” real
  - Determine the paths of NPCs
  - Make decision for NPCs based on players’ choices
  - Determine the strategy of play
  - Sense the behavior of player
  - Communicate and interact with player

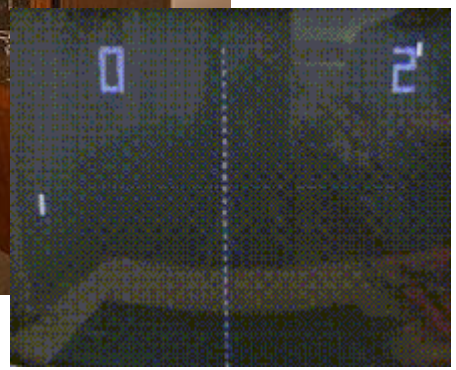
Goal of Game AI

- Be fun!
- Be challenging but not overwhelming (unless the hardest difficulty is selected)
- Avoid unfair advantages unless explicitly part of the game design; if used, disclose or balance them
- Need to follow the rules of the game
- Do not hog all the resources, such as CPU time

The Start of Game AI

- The video games developed in the 1960s and early 1970s, like Spacewar!, Pong (1972), Gotcha, were strictly based on the competition of two players, without AI
- Games that features a single player modes with NPCs started appearing in the 1970s such as the Taito game Speed Race (racing video game) (1974)
- It was during the golden age of video arcade games that the idea of AI opponents was largely popularized, due to the success of Space Invaders (1978)



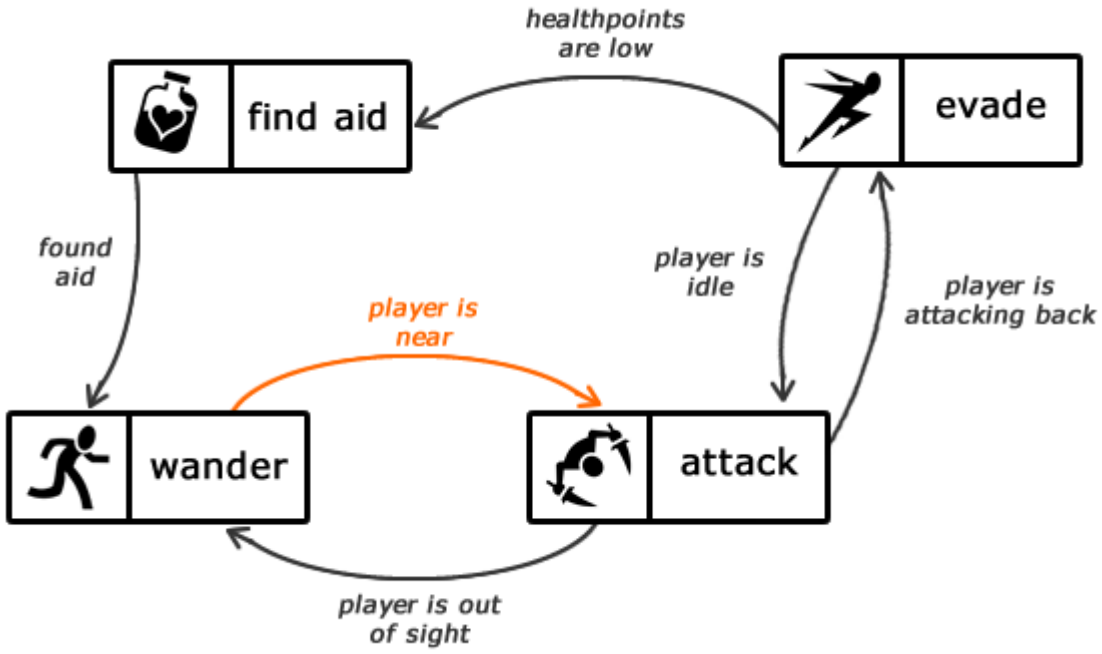






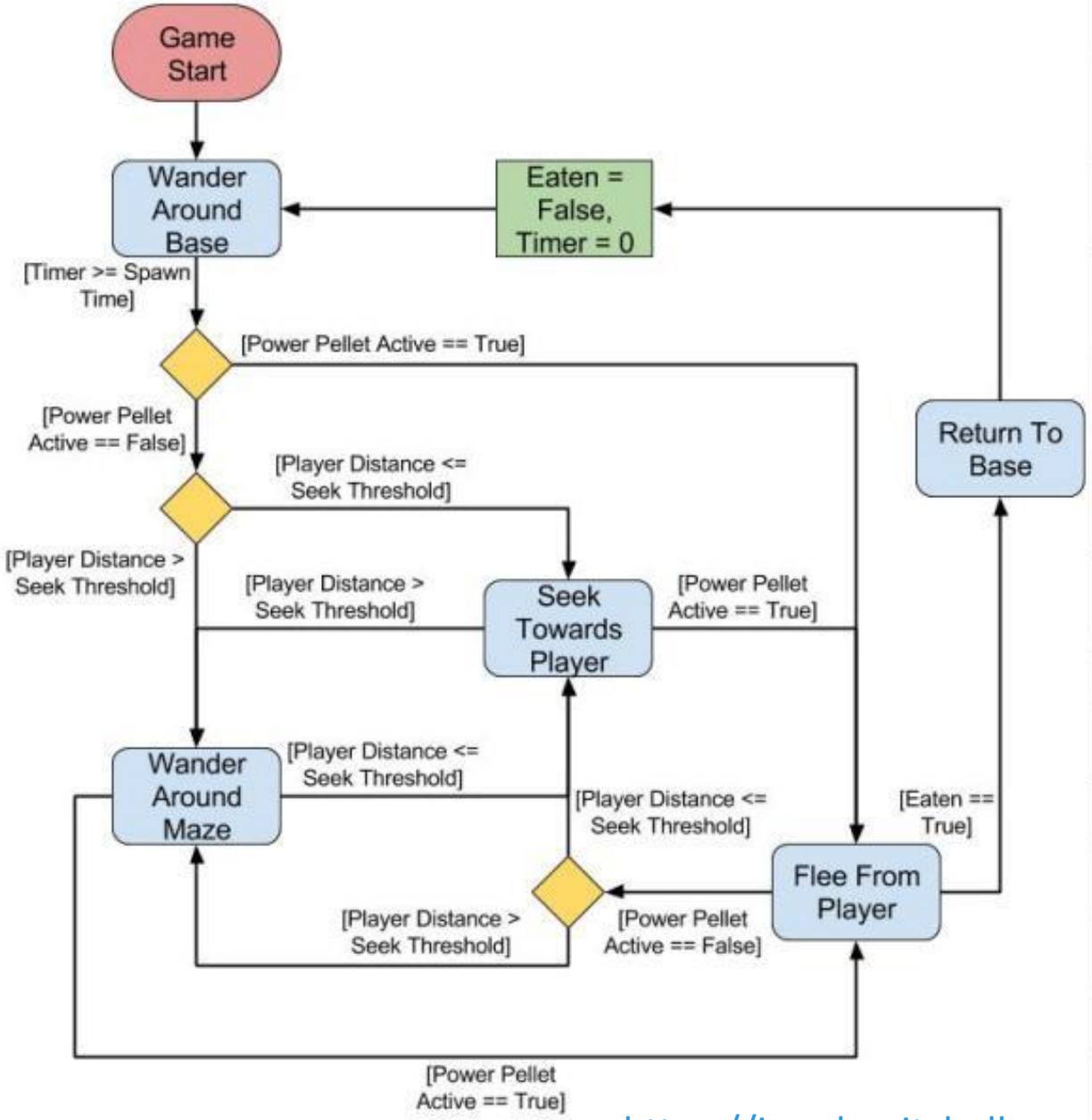
Finite State Machines (FSM)

- Early game AI was implemented with a rule-based system called Finite State Machine
  - Simplest and most basic AI model
  - Most common for Action games
- Four components
  - States which define behavior
  - State transitions which are the movement from one state to another
  - Conditions which must be met for state transition
  - Events/actions which are internally or externally generated which may lead to state transition

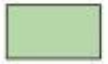


FSM example - Pac Man





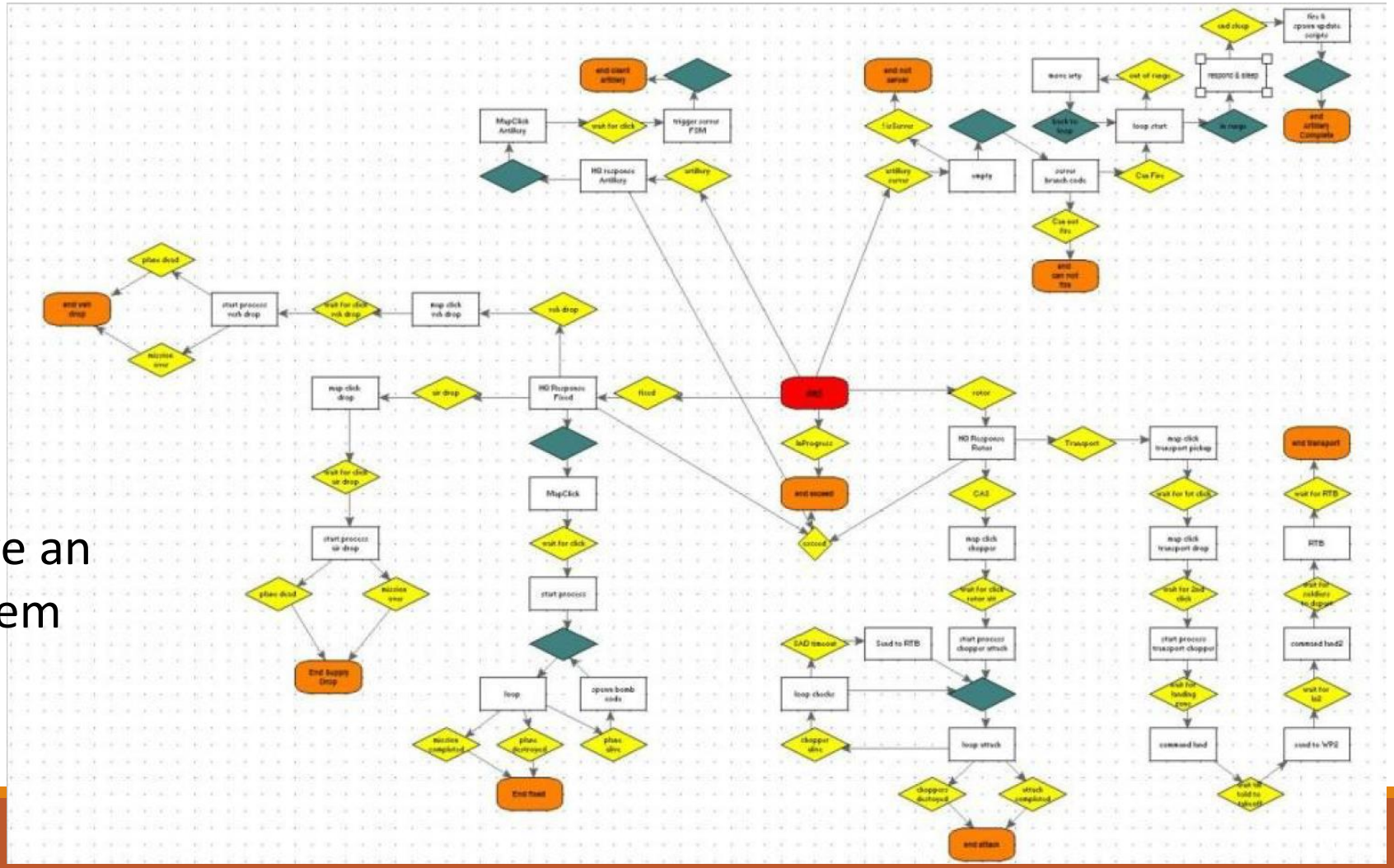
Variable Name	Variable Definition
Spawn Time	The amount of time a ghost must wait before exiting the base
Timer	A timer on the ghost that is compared to Spawn Time
Power Pellet Active	Whether or not the player has just eaten a Power Pellet
Player Distance	The distance between the player and the current ghost
Seek Threshold	The maximum distance from the player the ghost can be to seek the player
Eaten	Whether or not a ghost has been eaten

Symbol	Definition
	Terminal
	State
	Decision
	Process
	Transition
[asdfx]	Condition

https://jaredemitchell.com/ai-examples/

Finite State Machines (FSM)

- Modern FSM example



States can have an FSM within them

Finite State Machines (FSM)

- FSM disadvantages:
  - Behavior of NPCs is relatively predictable
  - Suitable only to simple games. State diagram becomes too complex to handle if there are many states
  - Since there are such crisp rules between states, NPC does not feel natural

Machine Learning in Video Game

- Machine learning (ML) is a subfield of AI where computer algorithms try to improve automatically through experience and using data
- These algorithms build and train a model using statistical analysis on the given dataset
- Make predictions or decisions without being explicitly programmed to do so
  - On contrary to FSM that the intelligence comes from the designer
- ML is used in video games for the following purposes:
  - Design smarter NPCs
  - Enhance computer graphics
  - Dialog generation and realistic interaction
  - Story and world generation
  - Create immersive games

Can a Game Agent Learn by Evolution?

- Imagine creating many game agents with different “brains.”
- Each agent plays the game and receives a score.
- The better agents are selected to create the next generation.
- Their “DNA” is a list of numbers that controls how the agent behaves.
- By repeating this process, agents gradually learn to play better.

Play game → Score agents → Keep the best → Create new agents

- Evolutionary Robots (ANN-based): https://www.youtube.com/watch?v=VUddPhXdBnY
- Flappy Bird with Neural Network & Genetic Algorithm: https://www.youtube.com/watch?v=aeWmdojEJf0
- Super Mario Bros. with Neural Network & Genetic Algorithm: https://www.youtube.com/watch?v=05rEefXlmhI

Design Smarter NPCs

- Machine learning makes NPCs less predictable and thus more realistic
- Can even possess skills beyond human
- E.g., Chess (Go) engine - AlphaGo Zero



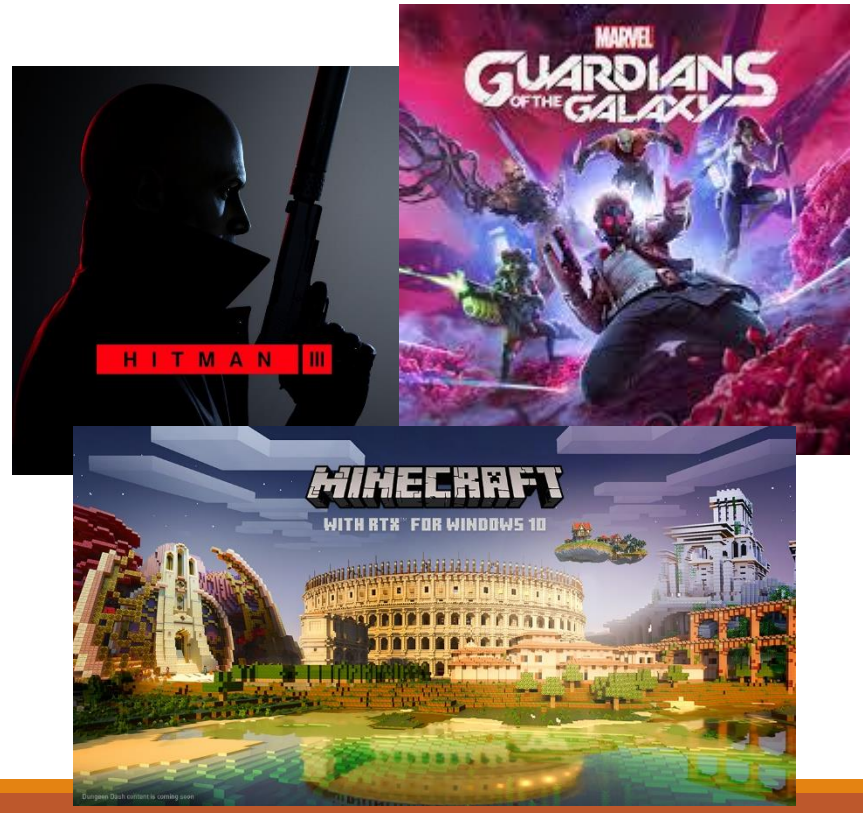
  - Google ‘s subsidiary DeepMind developed AlphaGo, which defeated Lee Sedol in 2016.
  - In 2017, DeepMind introduced AlphaGo Zero, which did not need human data to learn but only played against itself based on reinforcement learning
  - AlphaGo Zero (trained 40 days) has a play record of 100:0 with AlphaGo Lee (the one played with Lee Sedol)
  - Many professional Go players imitate the moves of AlphaGo and its successors

Enhance Computer Graphics

- Game players demand high fidelity graphics and smooth video
- Traditional graphics enhancement methods are often too time consuming or require extremely expensive hardware
- Through training with huge dataset, modern AI techniques, such as deep neural networks (DNN), can generate extremely realistic graphics
- Due to the massive parallel structure of DNN, they can be effectively implemented by Graphics Processing Unit (GPU) and can operate very fast
  - The same graphic enhancement task solved by DNN on GPU can be thousand times faster than using a traditional software program run on the computer's central processing unit (CPU)

Enhance Computer Graphics

- Example: Nvidia's Deep Learning Super Sampling (DLSS)
  - Traditional game engine often needs to reduce the frame rate if it is required to render high resolution graphics
  - Nvidia develops a post-processing technique called DLSS to solve the problem
  - Allows the game engine to render lower resolution graphics (so higher frame rate), then up-sample the graphics to the required resolution using DNN
    - Enabled by Nvidia RTX technology, extremely fast
  - Many games have the DLSS mode. DLSS requires a compatible NVIDIA RTX GPU, a supported game, and up-to-date drivers



Enhance Computer Graphics

- Minecraft with RTX
  - With DLSS, more than double frames per second (FPS) is achieved at the same resolution (4K) without sacrificing the graphics quality



https://www.youtube.com/watch?v=xyJxRRc3Pa0

Dialog Generation and Realistic Interaction

- Besides actions, machine learning can also be used to improve the gaming experience by formulating more accurate and realistic NPC responses
- Role playing games (RPG) make use of the dialog mechanism, which are significantly improved with the help of Natural Language Processing and sentiment analysis techniques using ML algorithms
- Example: Façade – real-time interactive drama
  - AI-powered game that allows the player to interactive with the NPCs in the game by natural language
  - Change the ending of the drama based on player's interaction
  - Natural language processing to understand the player's text input
  - Behavior language system to tell the NPCs what to response



Dialog Generation and Realistic Interaction

- Example: Façade – real-time interactive drama
  - In this example, a guy called Sergio (the player) visits the NPCs - Trip and Grace.
  - See the response of the NPCs when the player enters the text input



https://www.youtube.com/results?search_query=facade

Story and World Generation

- In Façade, the drama ending is automatically generated by a drama generator algorithmically based on player interaction
  - Can still trace the design of the programmers
- Game story can also be generated using machine learning
  - Can thus be totally based on the input of the player
- Example: AI Dungeon
  - AI Dungeon is a text-based adventure game that uses artificial intelligence to generate unlimited content
  - Rather than using a complex decision tree to script a large number of paths through the game, AI Dungeon uses a large language model to generate story content dynamically as a response to the players' typed prompts using natural language processing (NLP)



Story and World Generation

- Not only stories, game worlds can also be created by machine learning
- A number of popular games like Minecraft and the Grand Theft Auto series make use of an open world gameplay scenario
- Game world will be automatically generated while playing the game



- Example: Minecraft
  - Traditional games often include all their game world in the program. That makes their program extremely big
  - Minecraft can generate an astronomically large number of distinct worlds from world seeds, so only the generation algorithm and seed need to be stored.
  - It is because they are generated on-the-fly
  - In the future, game generation can be done by machine learning to make it adapt to the player's interest

Creating Immersive Games

- Video game developers are constantly trying to get games to be more immersive and realistic
- Modelling the real world is incredibly difficult
- A machine learning algorithm could help with predicting the downstream effects of a player's actions or even modelling things the player can't control, like the tiny motions of the players in a football match



- Example: FIFA 22 – the beginning of machine learning taking over animation
  - ML algorithm learns from more than 8.7 million frames of match capture, then writes new animations in real-time across a variety of interactions on the pitch
  - New Tactical AI to let each individual NPC player to better understand both attacking and defensive formations and movement

Creating Immersive Games

- Example: FIFA 22 – game play realism
  - Previous AI often picks 2 to 3 players in a group to animate. Some NPCs will thus make weird animation not relevant to the game play
  - FIFA 22 claims to achieve 11v11 animation through the HyperMotion technology that takes care of all the players in the field



https://www.youtube.com/watch?v=I9apQiMsJOU